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Bone scintiscanning in Orthopedics

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Introduction

Radionuclide imaging is perhaps the only means to image the physiology of skeletal system. About 30% of mineral loss must take place before an abnormality become visible on conventional radiograph [1]. This may take a few days to several months depending on the type of pathology. In contrast skeletal scintigraphy shows "hot spots" based on regional osteoblastic activity and becomes manifest within 24 hours of onset of insult. About 5 to 15% increase in bone turnover activity will show as a hot spot on a bone scan [2].

Technetium 99m-methylene diphosphonate is the radiopharmaceutical compound that optimally localizes to the bones 3 hours post intravenous administration. Images are acquired using a gamma camera that senses the low energy gamma radiation.

Bone scan is by far the most challenging of all nuclear scans to interpret because of its non specificity. Any pathology that accounts for variation in osteoblastic activity of the skeleton will result in increased tracer localization in the skeleton. This could be because of local or systemic cause. This is a pictorial essay that illustrates certain important entities encountered in day to day orthopedic practice.

Stress fracture/ sports injuries:

Shaft of tibia, fibula, metatarsal bone are prone to high impact stress injuries in athletes that vary in severity from simple bone bruise, shin splints to stress fracture. Bone bruise shows as mild to moderate uptake of tracer in the involved bone with normal 1st and 2nd phases. Shin splint appears as linear increased tracer localization in the shaft of tibia because of repeated high intensity exercise (Fig 1).

Stress fracture is a more severe form of injury that requires intensive care in terms of management and shows abnormality in all the three phases of bone scan. History is crucial to make a diagnosis (Fig 2).

When you have the right diagnosis, there is no differential diagnosis:

This is what our teacher used to say on the examination rounds. The saying is so true in day to day practice. As stated earlier, bone scan is quite non specific and exceptions prove the rule. Regional pain that aggravates at night in a young patient is quite typical of osteoid osteoma. Osteoid osteoma and fibrous dysplasia are two entities where the role of bone scan perhaps surpasses even that of histopathology [3,4].

Osteoid osteoma appears as focal avid tracer accumulation in the bone with a zone of less avid surrounding activity (double density sign). Radiologically there may be an area of sclerosis in the bone. Osteoid osteoma may be cortical, medullary, intracapsular, periosteal. Periosteal type of lesion may sometimes lead to significant sclerosis in the adjoining bone making diagnosis difficult on Plain radiograph, CT or MRI. However, bone scan will show a well defined focus of intense tracer accumulation in the region of nidus (Fig 3). The nidus appears as a small osteolytic area in CT scan.

Fibrous dysplasia appears as increased radiotracer accumulation in the involved bones. When multiple bones are involved the abnormality is usually confined to one side of skeleton. A typical scan finding obviates the need for histology (Fig 5).

Monostotic versus polyostotic lesion:

Another important indication of skeletal scintiscanning is to document involvement of multiple bones. Primary bone tumors such as osteogenic sarcoma, myeloma may show involvement of multiple skeletal sites. Benign entities such as fibrous dysplasia may also be polyostotic [4].

Systemic conditions with tell tale features on Bone scan:

Ill localized skeletal pain, bone aches and pains etc are a common manifestation of osteomalacia. The entity is often missed on clinical examination for lack of localizing signs. Some of these patients may also land up with insufficiency fractures.

Elderly patients often develop metabolic bone disease because of lack of sunlight exposure and vitamin D deficiency. Typical findings are diffuse increased radiotracer localization to the axial and appendicular skeleton with hot spots (representing loosers zone) in the medial cortex of proximal femur, ribs etc.

Primary hyperparathyroidism a disorder of bones, psychic moans, abdominal groans, renal stones and "ions" shows typical scan findings that are similar to those seen in osteomalacia (ie secondary hyperparathyroidism) Nuclear

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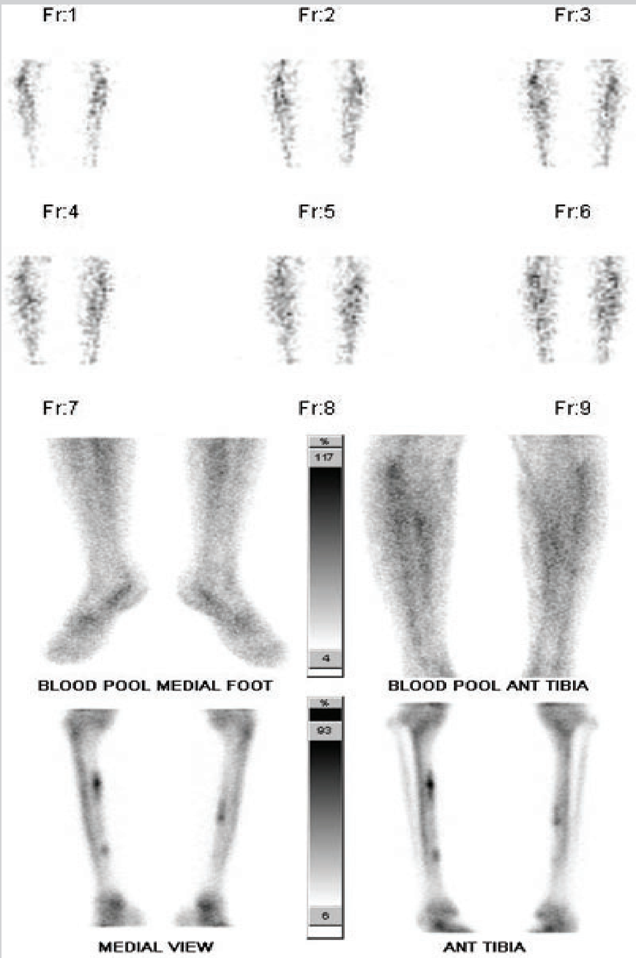

Figure 1: 30 yr old woman with recent onset exercise, scan shows typical shin splints.

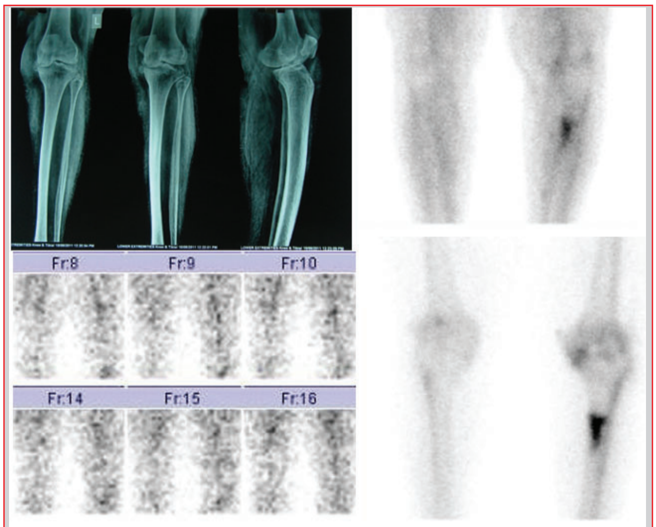

Figure 2: Bone scan shows Stress fracture in proximal shaft of left tibia.Xray-normal

medicine provides a lead in documenting parathyroid adenoma by technetium-Sestamibi scan or thallium-techentium subtraction scan. Technetium -99m localizes to the thyroid gland whereas thallium 201-chloride localizes to the parathyroid gland. Subtraction of the two helps in defining the parathyroid adenoma [5].

Hemolytic anaemia especially sickle cell type is known for vaso-occlusive crisis. Recurrent bone infarcts account for regional acute pain without any local signs of inflammation. Typically the first two phases of bone scan are normal, and the third phase shows diffuse cortical tracer localization. Relatively older patients also show tracer visualization in the spleen because of recurrent splenic infarcts a finding that is almost diagnostic of this condition (Fig 4).

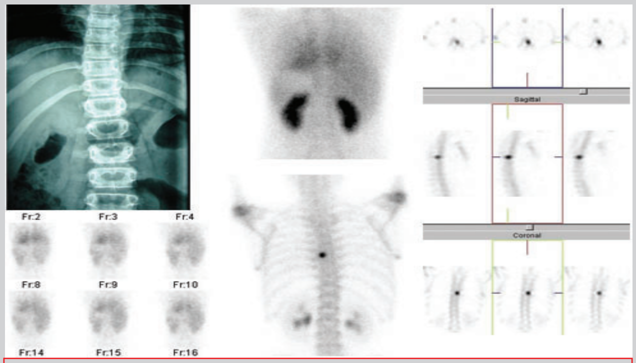

Figure 3: Bone scan shows focal avid tracer uptake in D7 vertebra. PlainXray shows scoliosis and SPECT images localize the osteoid Osteoma lesion in posterior element.

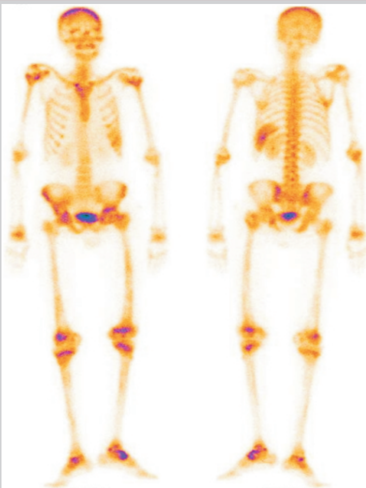

Figure 4: 24 yr old man with multiple skeletal symptoms and anaemia. Bone scan shows multiple cortical infarcts. Visualization of spleen is typical of Sickle cell disease.

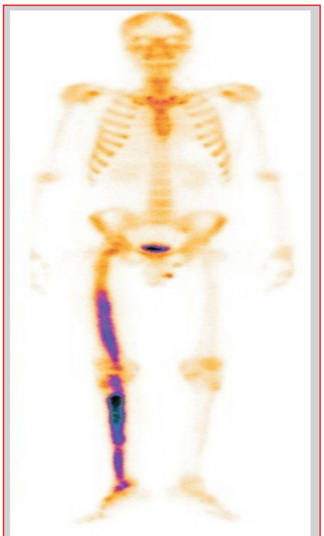

Figure 5: Polyostotic fibrous dysplasia.

Figure 6: Incidental diagnosis of Pagets disease of left hemipelvis.

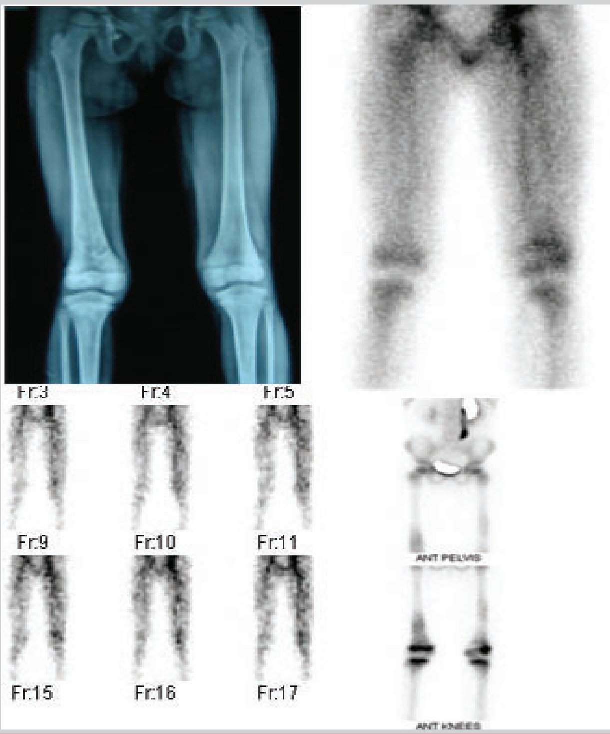

Figure 7: Inability to bear weight, this 4yr old girl was subjected to three phase bone scan to exclude osteomyelitis. Scan shows multiple bone infarcts; final diagnosis was acute myeloid leukemia.

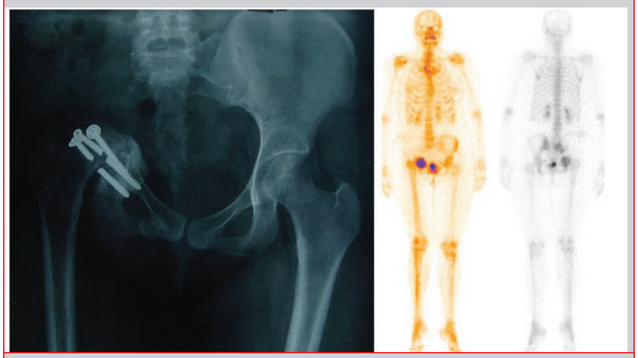

Figure 8: Osteogenic sarcoma right acetabulum with pathological fracture on Plain Xray Bone scan shows additional lesion in sacrum.

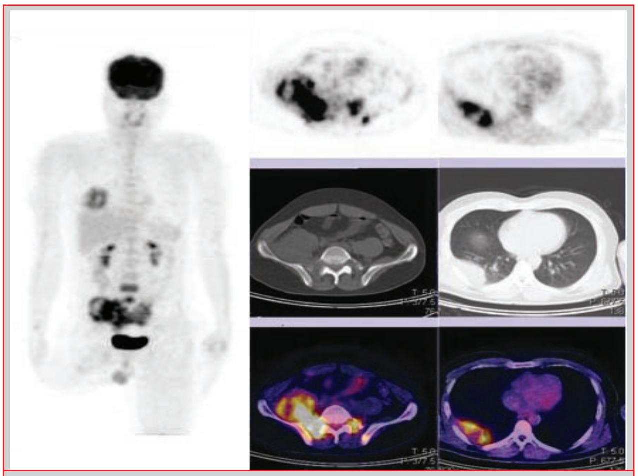

Figure 9: FDG-PET CT scan-Osteosarcoma neoadjuvant chemotherapy followed by surgery PET shows metastasis in pelvic bones and right lung.

Primary bone tumors[4]:

The purpose of skeletal scintigraphy is to look for evidence of involvement of distant skeleton in malignant neoplasms such as osteogenic sarcoma (Fig 8,9).

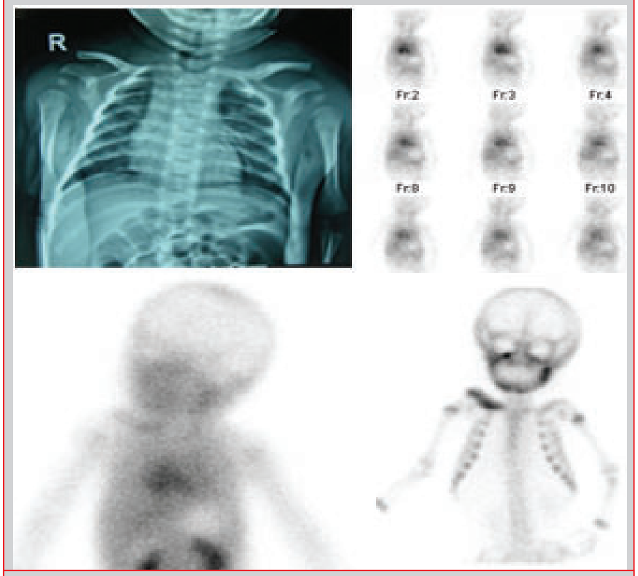

Figure 10: presenting with fever and restricted right shoulder movements this 3 month old child had normal radiograph. Bone scan shows osteomyelitis right clavicle.

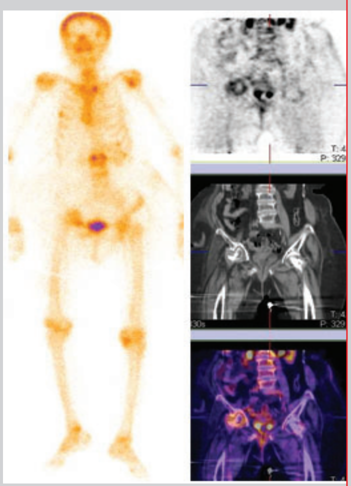

Figure 11: 73 yr old with right hip pain, bone scan versus FDG PET scan demonstrate incongruent uptake of FDG in right hip as compared to the bone scan compatible with infection.

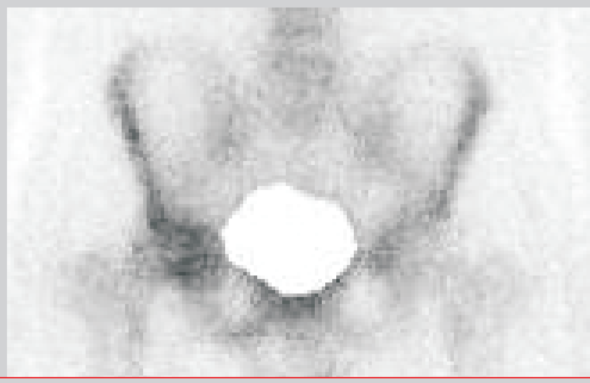

Figure 12: 23 year old with six month old hip dislocation shows no evidence of avascular necrosis of femoral head, bone contusion is evident from diffuse mild increased uptake.

Benign entities such as unicameral bone cyst, fibrous cortical defect evoke little osteoblastic activity and a small lesion may not be appreciated in absence of previous radiograph.

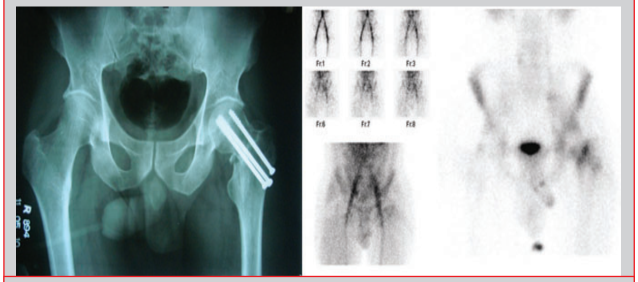

Figure 13: Left hip pain post fracture fixation shows well defined cold area in the head of left femur indicating avascular necrosis.

Bone infections [6]:

In a non traumatized skeleton, a three phase bone scan is a very sensitive means to document acute osteomyelitis. All the three phases show intense abnormality. Specificity is limited as malignant neoplasms may also look similar (Fig 10,11).

However in a previously traumatized bone such as prosthesis, old fracture etc, the specificity drops as trauma per se results in reactionary increased osteoblastic activity that may persist for months to as long as two years. Hence help is sought from additional radiopharmaceutical compounds with higher specificity for infection. These are Gallium-67 citrate, Technetium-99m- Leukocyte, Fluorine-18-fluoro-deoxyglucose PET/CT scan ((F18-FDG positron emission tomography/computed tomography).

Regions of skeleton with previous trauma and less intense uptake as compared to the bone scan/ normal uptake using the later compounds (eg Gallium, FDG PET) exclude infection. Infection appears as either congruent more intense tracer uptake of these compounds or incongruent uptake (ie in areas that do not appear hot on a bone scan).

PET/CT scan is also useful to stage osteogenic sarcoma and other malignant tumors. This test maps the whole body glucose metabolism. Neoplasms shows excessive glucose metabolism that precedes structural changes encountered on conventional imaging.

Avascular necrosis [7]:

Hip pain in patients with ethanol abuse, steroid exposure or following trauma is often related to avascular necrosis of head of femur. Bone scan shows cold area in the head of femur that is typical of this entity (Fig 12, 13). Hip implants do not interfere with scan interpretation.

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